

Autoresearch Loops

A Causal Study of the Instruction File

Christian Block

Research project with Prof. Ramesh

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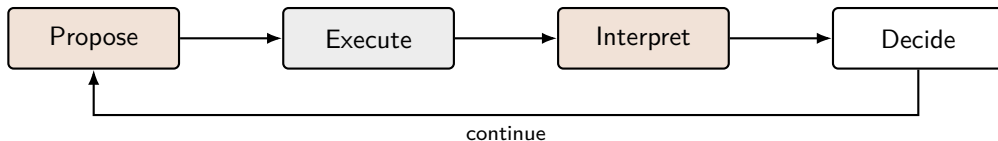
Manual ML research does not scale

- Tuning is search, run by hand
- One person, one experiment
- Overnight budgets sit unused
- March 2026: autoresearch automates it
- Forked over thirteen thousand times

The shift

The language model stops advising on experiments and starts running them.

A control loop, not a chatbot



- Same shape as Kubernetes reconciliation
- Observe, decide, act, repeat
- Shaded steps are model calls
- The executor is deterministic

The architecture is three files

- `prepare.py` — data and score
- `train.py` — model and training
- `program.md` — the instructions
- No scheduler, no driver program

Ownership

nobody edits `prepare.py`
the agent edits `train.py`
the human edits `program.md`

Consequence

`program.md` is not configuration handed to a controller. It is the controller.

It already works. That is the problem.

- Shopify Liquid, pull request 2056
- **53%** faster parse and render
- About 120 experiments, 93 commits
- Author called it probably overfitted
- Never merged

The gap

Benchmarks report whether a pipeline succeeded. None report which instruction was responsible.

Research questions

RQ1

Which components of `program.md` causally determine which aspects of the loop's behaviour, and how large is each effect?

RQ2

Do the components act independently of one another?

- Confounding blocks a direct answer
- Reworded files differ in every sentence

From prose to configuration vector

- One template, five variable slots
- Fixed part byte-identical everywhere
- Each slot written at two levels
- Nothing else in the file moves

A file becomes a vector

$$c = (c_1, \dots, c_5), \quad c_i \in \{0, 1\} \quad \implies \quad 2^5 = 32 \text{ files}$$

Choice of levels: the independent variables

	Component	Level 0	Level 1
<i>M</i>	Evaluation criterion	success undefined	5-fold CV accuracy
<i>B</i>	Search breadth	one model family	both families
<i>S</i>	Stopping rule	fixed 3 experiments	adaptive, up to 8
<i>O</i>	Reporting style	terse	full sentences
<i>E</i>	Search emphasis	explore first	exploit first

Two controls on the treatment

A second model wrote all ten paragraphs from fixed meta-prompts, then they were frozen. The two levels of a slot match to within ten per cent in token count.

Evaluation metrics: the dependent variables

Metric	Definition
Gain over default	recommended score minus $a_0(d)$
Wasted trial ratio	(rejected + duplicate) / attempts
Improvement rate	share of trials beating the best so far
Cost to best	executor calls to reach the peak

Why these, and not judge-based scores

Each is a difference in accuracy, a proportion of proposals, or a count of calls, read straight off the structured transcript. No scoring library and no judge model enters the measurement, so no metric can drift with the instruction being tested.

Design: run the whole space

Budget

$$N = \underbrace{2^5}_{\text{configurations}} \times \underbrace{D}_{2 \text{ tasks}} \times \underbrace{R}_{20 \text{ replicates}} = 1,280 \text{ runs}$$

- Configurations built, not collected
- Confounding removed by design
- Every interaction separately estimable
- A fraction would assume RQ2's answer

Controls

- Pinned model snapshot
- One stratified 70/30 split
- Same executor and whitelist
- Same seed list per cell

Temperature = 0.7

Not zero, deliberately. Run-to-run variance is the error term. Greedy decoding would zero it, and every standard error with it.

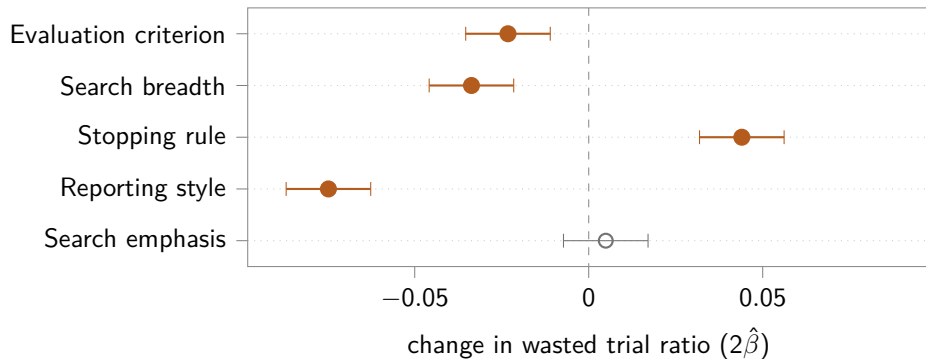
The regression model

Main effects, with the task controlled

$$Y = \beta_0 + \sum_{i=1}^5 \beta_i X_i + \sum_{j=2}^D \gamma_j D_j + \varepsilon$$

- Y : one of four metrics
- X_i : component i , ± 1 coded
- D_j : dataset indicator, the control
- γ_j absorbs task difficulty
- HC3 errors, Benjamini–Hochberg

What moved: the wasted trial ratio



filled = survives Benjamini-Hochberg bars = 95% CI $R^2 = 0.16$

Results in one slide

- Reporting style dominates: -0.075
- Wasted trials $10.6\% \rightarrow 3.1\%$
- Stopping rule: only outcome effect
- Cost to best: nothing survives
- Three of forty interactions significant

The task outweighs the instruction

On the improvement rate and the cost to best, the dataset effect is about twice the largest component effect, both at $p < 10^{-5}$. That is the variance γ_j removes from the residual.

Boundary conditions

- One agent model, one policy
- Semantics fixed, phrasing not varied
- Two datasets, one contrast
- Hyperparameter space, not code

Scope of the claim

A proof of concept. It establishes that editing the constraints in `program.md` has a measurable causal influence on the loop's behaviour; sizing that influence is left to wider designs.

Conclusion

- Instructions govern process, not outcome
- Verbose reporting: two thirds less waste
- Components act independently
- Method transfers to any sliced prompt

Why outcome barely moved

Untuned defaults already reach $a_0 = 0.960$ on both tasks and only 40% of runs improve on that. The ceiling binds, not the loop.

Thank you. Questions welcome.